

Two Connectivity Datasets, Two Different Worlds

Our `connectedness_hard` vs. Yehudai et al. 2025 — what each one hides from the transformer

diploma · 2026-06-20

Both datasets ask the **same yes/no question** — *is this graph connected?* — yet one is solved by a one-layer transformer at 100% and the other defeats it. This document compares them from every angle: how the graphs are built, how they look, what statistics they carry, how they become tokens, how those tokens are embedded, and — the crux — **what information each dataset leaks to the model and what it hides**.

1 The two tasks at a glance

	Our <code>connectedness_hard</code>	Yehudai <code>connectivity</code>
Graph size	variable, <code>n ∈ 12–24</code>	fixed, <code>n = 50</code> (every graph)
Construction	two dense blobs, hand-matched	gnp / rgg / scale-free / sbm generators
Connected class	the two blobs + 1 bridge edge	1 component (whole graph)
Disconnected class	the two blobs + 1 intra -blob edge	2–3 components, any sizes
Node features	normalised degree (1 value)	constant <code>x=1</code> (no signal)
Balance	500 / 500	2500 / 2500

The designs encode opposite **philosophies**. Yehudai samples graphs from natural random-graph models and lets the two classes differ however they naturally do. We **engineer** the two classes to be statistically identical in everything except the one fact that defines the label.

2 What each dataset looks like

2.1 Our data: two matched blobs, one edge apart

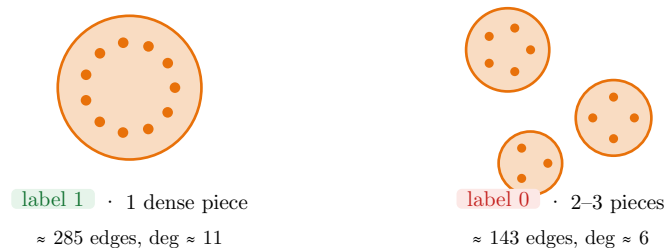
Every graph is two internally-dense blobs (each built as a Hamiltonian cycle + random chords, so every node has degree ≥ 2 — **no isolated nodes ever**). The label is decided by a single edge:



The two classes have the **same number of edges** (we add exactly one edge to each), the **same degree distribution**, and the **same blob structure**. The connected one routes its extra edge **between** blobs; the disconnected one routes it **inside** a blob. Spot-the-difference at the level of the whole graph.

2.2 Yehudai's data: natural graphs that differ in bulk

Their connected graphs are drawn from dense random-graph models; their disconnected graphs are a **union of 2–3 independent** such graphs. Nothing is matched — the classes differ in whatever way the generators make them differ.



Because a disconnected graph is several **smaller** pieces, it ends up with **far fewer edges** and **lower average degree** than a single connected graph of the same node count. That difference is not hidden — it is broadcast in every bulk statistic of the graph.

3 The statistics — measured, not assumed

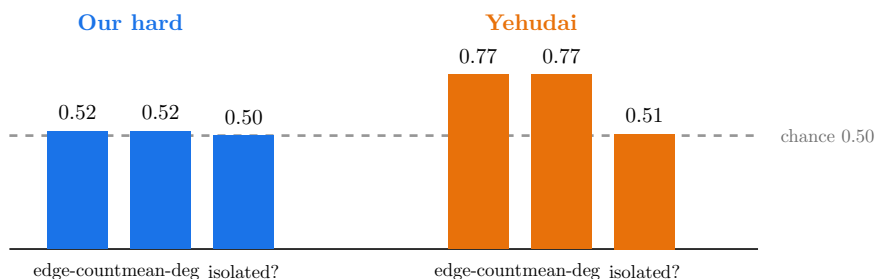
All numbers below are measured on the actual cached datasets (1000 of ours, 4000 of theirs).

Statistic	Our conn.	Our disc.	Yeh. conn.	Yeh. disc.
mean #edges	39.7	40.6	285.0	143.0
mean degree	4.49	4.52	11.40	5.72
min degree	2	2	1	0
mean #components	1.00	2.00	1.00	3.10

Read the columns in pairs. **Ours** (cols 2-3): connected and disconnected are nearly identical on every bulk statistic — $39.7 \approx 40.6$ edges, $4.49 \approx 4.52$ degree, both min-degree 2. **Theirs** (cols 4-5): connected has **double** the edges and **double** the degree of disconnected, and disconnected graphs contain isolated nodes (min degree 0).

3.1 How far a single number gets you

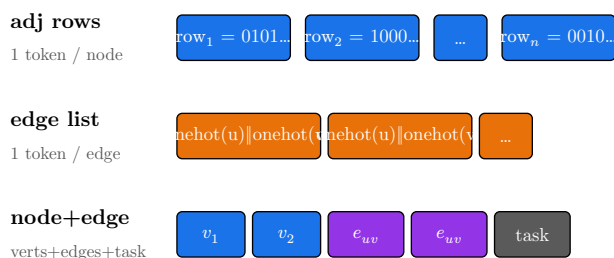
For each dataset, take the best possible **single-threshold** classifier on one bulk statistic:



A single scalar — total edge count, or mean degree — already classifies **77%** of Yehudai's graphs, but only **52%** of ours (chance). Our matched construction drives every cheap statistic to the coin-flip line; theirs leaves the answer lying in plain sight.

4 How a graph becomes tokens

A transformer reads a list of vectors. Three ways to flatten a graph into tokens are relevant here; all pad to the max node count.



- **adj rows** — token i is node i 's row of the adjacency matrix. Yehudai's winning choice.

- **edge list** — token per edge, each the concatenated one-hot identities of its two endpoints.
- **node+edge** — our Sanford-style sequence: vertex tokens, edge tokens, and a task token.

5 How tokens get embedded — and why it matters

After tokenizing, the model applies a linear embedding to each token and (for `adj_rows`) **mean-pools** over the node tokens before the classifier:

$$\text{graph vector} = \frac{1}{n} \sum_{i=1}^n W \cdot \text{row}_i = W \cdot \underbrace{\left(\frac{1}{n} \sum_i \text{row}_i \right)}_{\text{average adjacency row}}$$

Here is the punchline. The average adjacency row, summed, **is the mean degree**:

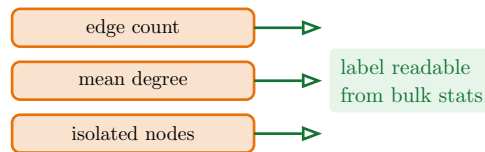
$$\sum_j \left(\frac{1}{n} \sum_i A_{ij} \right) = \frac{1}{n} \sum_i \underbrace{\left(\sum_j A_{ij} \right)}_{\text{degree of node } i} = \text{mean degree}.$$

So a single linear layer followed by mean-pooling can read out **mean degree directly** — no attention, no reasoning. On Yehudai’s data that scalar already separates the classes 77% (degree 11.4 vs 5.7), and a little attention closes the rest. The 1-layer `adj_rows` transformer isn’t **reasoning about connectivity** — it is **measuring how dense the graph is**.

On our data the same readout returns ≈ 4.5 for **both** classes. Mean-pooling gives the classifier nothing. To separate our classes the model must look at **which specific node connects to which** — it must actually trace whether a path crosses between the blobs. That is real global reasoning, and it is exactly what we wanted to force.

6 What each dataset hides from the transformer

Yehudai: the answer leaks out



Ours: only structure carries it



In one line. Yehudai’s dataset lets the model answer “connected?” by asking “how dense?”. Ours removes density, degree, and isolated-node cues entirely, so the only remaining signal is the actual reachability structure — which a position-based tokenization (adjacency rows, one-hot edge lists) cannot read off without solving the problem.

7 Why this matters for the thesis

The empirical chain (see `yehudai-empirical.md` and `tokenization.typ`):

1. **Our model is not the bottleneck.** Our adjacency-rows transformer scores **1.00** on Yehudai’s data (by epoch 4) — it reproduces their result exactly.
2. **The same model scores ≈ 0.6 – 0.7 on ours**, even with size fixed.

3. Therefore the gap is the **dataset**. Yehudai’s benchmark is separable by bulk statistics that flow straight through mean-pooling; ours is engineered so they don’t.

This makes `connectedness_hard` a **strictly harder probe of global reasoning**: it is the connectivity question with every cheap shortcut sealed off, leaving only the genuinely parallel/depth-bounded computation that the theory (Sanford et al. 2024a) is actually about.

Model · data	Tokenization	Test acc
our GNN · Yehudai connectivity	adj rows	1.00
our GNN · our hard (fixed n=20)	adj rows	≈ 0.70
our GNN · our hard (variable n)	adj rows	≈ 0.59
our transformer · our hard	node+edge	0.50 (ln 2 plateau)